

Design Solution	Description
HoverSlide (Magnetic Slider with pulley-handle elevation)	When the snack falls down, it enters the dispenser bin. As the user slides the handle to the right, the dispenser bin with the snack slides up with a string and pulley system. Magnetic levitation system is used to reduce force needed between the sliders. There is only one square opening in the slider for the snacks to go through, such that no matter where the opening is, it is either blocked by the dispenser bin or a solid surface within.
Snack n' Slide (Drawer)	A wheel driven or sliding drawer is put in place of the retrieval opening. The user can pull or push on the handle to open and close the drawer and access purchased products. A small flap with a one-way, sensitive, spring-loaded hinge is present to allow products to enter the drawer but provide a sufficient anti-theft functionality.
Scoopy Doo (Scoop)	The tool can be inserted into the retrieval bin and then maneuvered by the user such that the snack sits on top of the base. The tool then can be pulled up and out of the retrieval bin so the user can easily grab the product.
Snack Chute (Inner One-Way Flap)	The outer flap that the user needs to reach through is removed and is replaced by an open tray/bin that they can reach into to retrieve the snack. An inner flap that is parallel to the ground is added below the snack shelves, when the snack falls down as usual, the force of gravity will allow the snack to push the flap down and fall through into the retrieval bin. The flap is one way, with bars on the upper side preventing it from opening up.
Wave n' Take (Sensor Operated Flap)	Sensor is installed adjacent to the flap door and angled downwards. Once a valid payment is made, the user waves to activate the sensor which displays a green light and opens the flap after 7 seconds. The flap remains open for 45 seconds, then flashes a red light before closing slowly. This reduces the required force and maintains the original anti-pilfer system. A different material will be used for the flap door to reduce the friction between the anti-pilfer mechanism and flap door.
The Snack Sack (Plastic Pouch with inclined slope and higher maximum opening angle of flap)	The maximum opening angle of the flap is increased by reducing friction between flap and anti-theft flap, allowing the flap to slide further back. An inclined slope and plastic pouch is attached to inside of the machine that allows snacks drop onto the slope, slide down along it and held by the plastic pouch which is within reasonable retrieval range

Design Solution	Advantages	Disadvantages	Requirements
HoverSlide (Magnetic Slider with pulley-handle elevation)	1. Reduces required force for snack retrieval 2. Increases the height of the product retrieval bin to the optimal height according to EC 3. Preserves anti theft feature	1. Changes the vending machine by a significant amount 2. There is a protruding outer box attached to the machine, which can be exposed to hitting.	All requirements met
Snack n' Slide (Drawer)	1. Easily bring product of any size to user with minimal force 2. After one quick action, nothing dividing user and product 3. Has limited modifications of the existing machine 4. Reduces the reach depth needed to retrieve the product	1. Has a weak anti-theft system 2. Element that protrudes beyond the current vending machine 3. Does not fully remove inconvenience of bending down 4.	All requirements met
Scoopy Doo (Scoop)	1. No major alterations to current vending machine 2. Product of any size is brought to user 3. No need for excessive bending 4.	1. Does not significantly reduce force or time for product retrieval 2. Relies on the user being able to adequately use device 3. May require uncomfortable motions to get product onto base 4. May interfere with bystanders	All requirements met
Snack Chute (Inner One-Way Flap)	1. Reduces required force for snack retrieval 2. Increases the height of the product retrieval bin 3. Preserves anti theft feature	1. Does not fully remove the inconvenience of bending down 2. Dispensing process may damage the contents of the snack 3. If snack is not heavy enough, it may not fall through the flap 4. It is harder to implement	All requirements met
Wave n' Take (Sensor Operated Flap)	1. Reduces required force for snack retrieval 2. Maintains original anti-pilfer system 3. Increases the height of the product retrieval bin 4. Intuitive and easy use for user	1. Requires significant structural change to vending machine 2. Difficult implementation (several layers to the sensor) 3. Significant increase in complexity (maintainability, assembly) 4. High chance of malfunction (sensor triggered too early, etc.) 5. Still requires user to bend down and reach into retrieval bin	All requirements met
The Snack Sack (Plastic Pouch with inclined slope and higher maximum opening angle of flap)	1. Reduces required force for snack retrieval 2. Maintains original anti-pilfer system 3. Preserve integrity of original vending machine 4. Practical, easy to implement easy use for user		All requirements met

Evaluation Criteria	HoverSlide (Magnetic Slider with pulley-handle elevation)	Snack n' Slide (Drawer)	Scoopy Doo (Scoop)	Snack Chute (Inner One-Way Flap)	Wave n' Take (Sensor Operated Flap)	The Snack Sack (Plastic Pouch with inclined slope and higher maximum opening angle of flap)
The less force required to retrieve the snack, the better.						
The shorter the duration of force exertion (the shorter the retrieval process), the better.						
The less body part flexibility required to exert this force, the better.						
The smaller the reach depth, the better.			N/A			
The less compression or tension the product has to undergo, the better.						
The less body part flexibility needed to reach the entire depth, the better.						
The more natural and ergonomic the hand movement throughout the retrieval process, the better						

